



PHARMACEUTICAL INGREDIENTS MANUFACTURER

A global pioneer in pharmaceutical ingredients manufacturing faced challenges with outdated systems, haphazard lab data management, data accessibility problems and compliance issues.

To support growing regulatory demands and operational complexity, the company needed a modern, cloud-based solution to enhance efficiency, streamline data management, and ensure compliance with industry standards like 21 CFR Part 11.

Logilab SDMS was implemented to address these issues, thus marking a significant transformation in their data management processes, delivering enhanced efficiency and improved data security.

This implementation not only resolved the challenges posed by their legacy systems but also positioned the company for future growth by providing a scalable, cloud-based solution that can be optimized in accordance to the company's evolving needs.



WHAT'S UP WITH PHARMACEUTICAL LABORATORIES

A leading provider of pharmaceutical ingredients, with over 20 production sites and more than 100 years of industry expertise, has built a long-standing reputation for excellence. They play a pivotal role in supplying high-quality ingredients to the pharmaceutical, food, and nutrition sectors. These industries are heavily regulated and require rigorous precision and compliance in data management, especially as laboratory operations grow increasingly complex. Thus, a transition from a legacy system to a more modern and agile platform is necessary to address these demands.

Some of the pressing challenges included,

Outdated Architecture: The existing IT and Laboratory Information System (LIS) was hosted on internal servers, resulting in delays in data processing and reporting. The on-premises setup lacked seamless cross-site communication and hindered efficient data management across borders, reducing overall operational efficiency.

Redundant Systems: A previously implemented solution proved overly complex and was not optimized for cloud-based operations. The redundant functionalities and lack of intuitive workflows resulted in underutilization of the system and created inefficiencies.

Disorganized Data Management: With large volumes of data generated daily, a system was needed to securely store and intelligently organize information. An effective solution would archive files using key identifiers like batch numbers, sample IDs, and other critical metadata, while offering robust tagging and search features for quick, effortless data retrieval.

Migration Costs: Transitioning data from one platform to another is often fraught with challenges, and for the company, migrating data from its legacy system was proving both costly and time-consuming. The previous vendor's platform made the process complex, prompting the company to seek a solution that would enable smoother, more cost-effective migration without compromising data integrity or compliance.

Regulatory Compliance: Compliance with standards like 21 CFR Part 11 and ALCOA+ principles was non-negotiable. Their existing systems struggled to meet these regulatory demands, leading to concerns over audit readiness and overall compliance.

HOW LOGILAB SDMS BRINGS AGILITY TO SCIENTIFIC DATA HANDLING

Q: What if you had a cloud-based system that streamlined lab data management and enhanced data accessibility?

A: With full automation in the data archiving process, automated data parsing and tagging, and powerful search functionality, Logilab SDMS transforms lab data management by making data easily accessible, organized, and compliant.

GET THE PICTURE

The pharmaceutical ingredients manufacturing company's laboratories generate vast amounts of data daily, collected from a multitude of instruments involved in product testing and quality assurance processes. Managing and organizing this data is critical not only for operational efficiency but also for regulatory compliance.

Recognizing the limitations of its legacy data management systems, the company sought a modern, cloud-based solution to streamline its data handling, ensure compliance with global regulatory standards, and improve overall operational efficiency.

AUTOMATED PHARMACEUTICAL LABORATORIES

In response to these challenges, the pharmaceutical ingredients manufacturing company chose to implement [Logilab SDMS](#) (Scientific Data Management System). [Agaram Technologies](#) worked closely with the company to understand their data organization needs and tailored Logilab SDMS to meet their unique requirements, offering a flexible, cloud-based solution that integrated seamlessly with their existing infrastructure while addressing the key pain points of their legacy system.

CUSTOMIZATION FOR TAGGING & FOLDER STRUCTURE

The SDMS was customized to automatically read, archive, and tag data from laboratory instruments based on pre-configured file and folder structures that aligned with their operational workflows, such as grouping files by batch number, sample ID, and instrument type. This eliminated the need for manual data entry, reducing the chances of human error.

Additionally, the SDMS could parse data from within files, such as extracting critical values from PDF reports, making this data easily searchable. This functionality enabled users to quickly retrieve information without sifting through raw files, significantly improving data accessibility and usability.

DATA PARSING & SEARCH FUNCTIONALITY

Enhancements were made to the system's search functionality; allowing users to search not only by tags but also by parsed data within files such as key values extracted from instrument-generated reports. This feature enabled more granular searches, improving data retrieval.

A reverse search capability was introduced where users could select an instrument and immediately retrieve data related to that instrument, along with any associated clients. This Client-Instrument mapping feature drastically improved the accuracy and speed of historical data retrieval, an area that had been a major pain point in the past.

PHASED IMPLEMENTATION

Although the project was initially slated for completion by March 2024, internal change control processes at the company extended the timeline to June 2024. This phased approach ensured that all critical functionalities were thoroughly tested and validated before going live, minimizing the risk of operational disruptions.

KEY BENEFITS

- **Improved Efficiency:** The automation of data management processes significantly reduced the time spent on manual data handling, enabling faster data archiving and retrieval. The parsing and tagging features further enhanced the speed at which critical information could be accessed.

- **Cloud-Based Centralized Control:** With a centralized, cloud-based SDMS, the company could manage data from multiple global locations in real-time. This improved operational visibility and allowed for more effective decision-making, particularly in scenarios where quick access to data was essential.
- **Cost-Effective Migration:** The migration process was streamlined, avoiding the prohibitive costs associated with transitioning data from the previously implemented solution. Logilab SDMS offered a more efficient and affordable solution without sacrificing data integrity or functionality.
- **Enhanced Data Security and Integrity:** Logilab SDMS ensured that all data was securely stored in the cloud, protected by encryption and secure access controls. The system's compliance with 21 CFR Part 11 and ALCOA+ principles ensured that data was accurate, reliable, and traceable, fulfilling regulatory requirements.
- **Increased Search Accuracy:** The introduction of Client-Instrument mapping and reverse search functionalities improved the accuracy and speed of data retrieval, reducing the time spent searching for historical data.
- **Scalability and Future Readiness:** Logilab SDMS was designed with scalability in mind, capable of handling increasing data volumes as the company's operations grow. Future additions, such as blob storage, will further enhance the system's ability to manage large datasets.

INSIGHTS

- **Improved Search Tools:** The successful implementation highlighted the importance of advanced search functionalities, such as reverse search and data parsing, which greatly enhanced the user experience.
- **Security Enhancements:** The project underscored the need for robust security features, including secure FTP communications, which Agaram Technologies enhanced during the implementation process.
- **Scalable Architecture:** Moving to a cloud-based architecture proved to be a critical decision, allowing the company to efficiently scale its data management capabilities across global sites.

By implementing Logilab SDMS, the pharmaceutical ingredients manufacturing company unlocked cloud-based global access, streamlined operations, and ensured enhanced data accessibility and compliance to regulatory standards.

Curious to see the product in action? Request a demo and get a detailed overview [here](#).

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